Expert-in-the-Loop Systems for High-Quality Training Data Construction: A Survey

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***Abstract*—High-quality training data form the foundation of reliable intelligent systems in knowledge-intensive and safety-critical domains such as healthcare, intelligent transportation, smart manufacturing, cyber–physical systems, and scientific discovery. However, constructing such data still depends heavily on domain experts, whose nuanced judgment remains essential in tasks that require deep domain knowledge, contextual reasoning, and adherence to professional standards. To bridge the gap between expert insight and scalable data production, a growing body of research has developed the Expert-in-the-Loop (EITL) paradigm, which integrates human expertise with AI methods and collaborative platforms throughout the data life cycle. In this survey, we present a unified, systems-oriented analysis of EITL methodologies along two orthogonal dimensions: training-data types and expert–AI collaboration paradigms. We systematically categorize existing approaches according to expert roles and the augmentation mechanisms of AI systems, ranging from rule-based preliminary labeling and active sampling to generative synthesis and model-assisted evaluation and quality control. By abstracting common architectural patterns and interaction mechanisms, this survey establishes a structured taxonomy and conceptual framework for expert-guided data engineering. It further outlines open challenges and future directions for scalable, reliable, and human-centered data construction in contemporary AI development.**

***Index Terms*—Expert-in-the-loop systems, human–AI collaboration, expert-augmented intelligence, expert-embedded systems, data construction.**

I. Introduction

With the continuous advancement of large language models (LLMs), general-purpose language generation has expanded rapidly [1], while multimodal systems such as Flamingo and visual-instruction-tuned models have extended these capabilities to cross-modal tasks [2], [3]. A recent survey of data augmentation further highlights the growing importance of diversified training data across modalities and tasks [4]. At the same time, deployments in finance, healthcare, and law [5], biomedicine [6], and industrial time-series analysis [7] expose persistent requirements for domain knowledge, reliability, and explainability [8]. Medical foundation-model studies document both specialized opportunities and factual or regulatory risks [9]–[11], while domain-specific continual pretraining in law demonstrates that general pretraining alone does not ensure specialized performance [12].

Supervised data provide explicit input–output mappings that guide models in learning task semantics and decision rules. Domain-specific corpora and evaluation procedures have supported legal adaptation, medical reasoning, and learning from medical textbooks [13]–[15]. Scenario engineering with synthetic data offers another path to trustworthy domain adaptation [16], while broader analyses of AI-generated content clarify the opportunities and governance risks of automated generation [17]. These examples motivate a high-quality data ecosystem in which expert judgment is used where semantic validity and professional standards cannot be established automatically.

However, fully manual annotation remains time-consuming, costly, and difficult to scale. Quality-control studies also illustrate distinct operational problems: LLM-assisted cell-type annotation requires explicit reliability evaluation [18], while conflict-aware group decision-making requires mechanisms for structuring and resolving divergent feedback [19]. Automated annotation from natural-language specifications in autonomous driving [20] and cost-effective active semi-supervised learning with crowds [21] demonstrate complementary efforts to reduce labeling burden. Thus, improving annotation efficiency while preserving quality and enabling traceable conflict resolution remains a central challenge in scaling high-quality training data production.

Building on the limitations of fully manual and fully automated approaches, the Expert-in-the-Loop (EITL) paradigm has emerged as an increasingly important approach. It integrates experts as core participants throughout the data creation pipeline—spanning data sampling, annotation, validation, and iterative refinement—thereby enabling dynamic feedback, real-time correction, and task-driven guidance through continuous human–AI collaboration. Compared with fully automated or purely manual annotation, EITL can enhance data credibility and semantic accuracy while helping to identify and mitigate model biases and improve the relevance of data to target tasks. Within this framework, expert knowledge is transformed into actionable supervision signals that guide data generation, thereby supporting more efficient and precise fine-tuning of large models.

Despite these advances in data-construction paradigms, ex-

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isting reviews have primarily addressed data processing and data augmentation but lack a systematic synthesis of the methods and platforms used to generate high-quality training data with expert involvement. This survey makes four contributions: (i) it distinguishes EITL from general HITL and characterizes the evolution of expert roles; (ii) it develops an orthogonal taxonomy based on training-data type and AI agency; (iii) it connects data-construction methods with their platform implementations and illustrates their interaction through a case study; and (iv) it synthesizes evaluation dimensions, persistent challenges, and future research directions. Together, these contributions characterize the technical ecosystem of expert–AI collaborative data generation and provide conceptual and practical guidance for scalable, domain-specific production of high-quality training data.

Section II presents the background to this review, focusing on the definition of the EITL paradigm and the limitations of existing surveys. Section III proposes a unified typology that categorizes current expert–AI collaboration methods and platforms according to both the type of data being generated and the level of AI agency. Sections IV–VI analyze how intelligent generation methods and data engineering platforms address the challenges associated with each data type. Section VII presents a case study that illustrates the corresponding data-generation processes. Finally, Section VIII discusses current challenges and future directions for data generation under the EITL paradigm and provides a concluding synthesis.

II. Background

A. Expert-in-the-Loop (EITL)

Human-in-the-loop (HITL) [22] is a foundational paradigm that integrates human input throughout the training, refinement, evaluation, and interactive control of AI systems [23]. Its breadth includes assistive and rehabilitation robotics and human-in-the-loop teleoperation [24], [25]. In data construction, however, a crucial distinction remains: not all human input is equivalent. In knowledge-intensive domains, the involvement of *experts*—professionals with deep conceptual understanding and procedural expertise—is essential for ensuring both the accuracy and epistemic validity of AI-driven decisions.

This motivates the shift toward **Expert-in-the-Loop (EITL)**, a specialized form of HITL in which expert judgment shapes the semantics, structure, and reliability of training data. Unlike conventional labeling, EITL enables the construction of *knowledge-grounded datasets*, as shown in Fig. 1. Their annotations capture not only *what* is observed but also *why* it is significant, embedding causal reasoning, contextual nuance, and domain-specific uncertainty.

Consequently, methods and platforms designed for EITL require evaluation beyond standard performance metrics. They should be assessed from the perspective of domain experts, with attention to scientific soundness, practical usability, and model explainability. Only through such holistic analysis can we understand how AI tools can effectively augment expert cognition and ultimately improve the efficiency and integrity of knowledge-intensive data-curation pipelines.

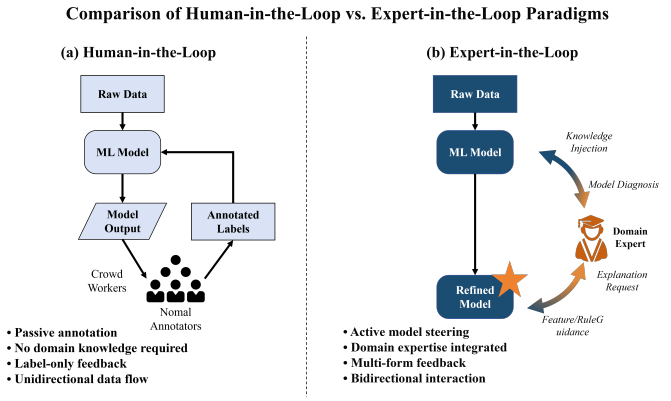


Fig. 1. Comparison of (a) Human-in-the-Loop and (b) Expert-in-the-Loop paradigms. The illustrated conventional Human-in-the-Loop annotation workflow relies on nonexpert crowd workers providing unidirectional, label-only feedback, whereas the Expert-in-the-Loop paradigm emphasizes bidirectional, knowledge-rich interaction between domain experts and the model, incorporating diverse forms of feedback such as domain knowledge injection, model diagnosis, error correction, and rule and constraint guidance.

B. Existing Surveys

Having established the importance of expert involvement, we note that data construction has received increasing attention as AI technologies have advanced. The performance of intelligent models is widely recognized as being fundamentally constrained by the scale, quality, and diversity of their training data. Recent surveys have examined training data from multiple perspectives, including data collection, human–computer interaction, and quality control. We organize these surveys into three broad stages:

Stage I: Traditional data construction. Neves et al. (2014) [26] summarized annotation tools and expert annotation workflows in the medical domain. Roh et al. (2021) [27] systematically discussed approaches such as crowdsourced annotation, semiautomatic labeling, and data augmentation in data collection, addressing issues including inter-annotator agreement and quality-control mechanisms. However, these works were published before the rise of foundation models and therefore did not cover AI-assisted data-construction scenarios driven by such models.

Stage II: Data construction through human–AI collaboration. Growing data-quality demands highlight the value of human–AI collaboration in constructing training datasets. Wu et al. (2022) [22] reviewed interactive data annotation, feature engineering, and model tuning, but data construction was discussed only as one component of the overall workflow and was not treated as an independent, systematized topic. Teso et al. (2023) [28] reviewed explanation-mediated interaction for learning, editing, and debugging models, emphasizing how explanations can support more effective human feedback.

Stage III: Human–AI collaborative data construction in the foundation-model era. Foundation-model technologies have intensified human–AI interaction, driving a shift toward data-centric AI. Zha et al. (2023) [29] systematically examined human–computer interaction issues in data quality assessment, data cleaning, and feature engineering, explicitly proposing a paradigm shift "from model-centric to data-centric." Long

et al. (2024) [30] provided a systematic survey of LLM-driven synthetic data generation, decomposing the generation pipeline into three stages: data generation, data curation, and data evaluation. Tan et al. (2024) [31] surveyed LLM-based annotation generation, assessment, and utilization, but did not organize the field around domain-expert roles or method–platform implementation. Liu et al. (2025) [32] presented a comprehensive review of LLM training data, covering the entire pipeline from traditional web crawling and public corpus integration to instruction-tuning data construction. More recently, Eckert et al. (2026) examined how human linguistic skills and responsibilities are changing as generative AI becomes embedded in data work [33]. In this stage, instruction and preference data became central forms of high-quality training data.

However, current discussions of human–AI collaborative data construction remain fragmented. To the best of our knowledge, existing surveys have not systematically examined domain experts as the core collaborative agents in data construction across the full life cycle, including the definition of data requirements, sample generation, quality validation, and iterative optimization.

Fig. 2 illustrates three broad stages in the evolution of AI data construction, transitioning from labor-intensive paradigms to automated and synthetic approaches. While this evolution has substantially improved efficiency, it has also introduced the risk of diminished domain specificity and expertise. In response, the gold-shaded section on the far right presents our core contribution: in the context of the foundation-model era, we provide a comprehensive review of end-to-end data collaboration mechanisms centered on domain experts. Complementing prior surveys that focus on data processing, general HITL, or LLM-based annotation, this work positions expert-role evolution as the central organizing principle, enabling a systematic analysis of how expert authority transitions across paradigms, from Pure Expert annotation to Expert-Guided AI governance. To the best of our knowledge, existing surveys have not adopted this framing as their primary organizing principle. The review therefore combines two complementary dimensions: (i) an orthogonal classification framework based on data types and levels of AI agency and (ii) a methodological and platform-oriented perspective that provides a holistic synthesis of existing approaches.

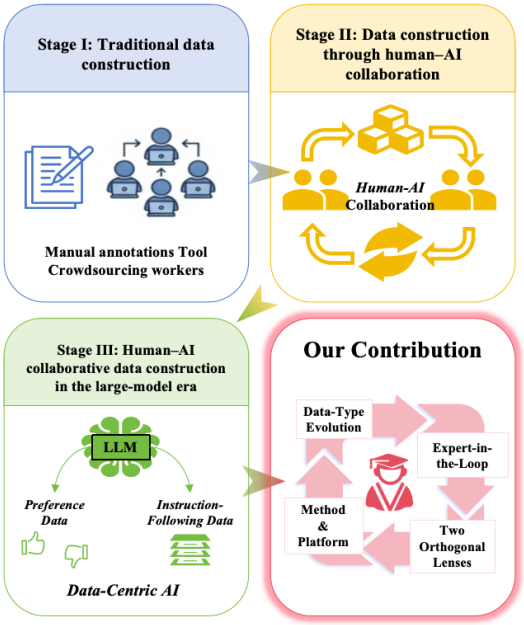


Fig. 2. Evolution of data construction paradigms in AI research.

III. Taxonomy

To systematically address the gaps identified in prior studies, we propose a unified taxonomy that categorizes methods used in or enabling EITL-based data generation according to the type of training data being produced and the expert–AI collaboration mechanism. We distinguish three core categories—supervised data, instruction-following data, and preference data—and analyze each across levels of AI agency in terms of methodological principles, platform requirements, and workflow patterns.

A. Overview of the Taxonomy

As shown in Fig. 3, we present a systematic taxonomy of methodologies for expert-involved dataset construction. The framework is structured along two orthogonal dimensions: **Data-Type Evolution**, representing the increasing abstraction of supervision signals, and **Increasing AI Agency**, illustrating the paradigm shift in human–AI collaboration. This taxonomy highlights the transition from labor-intensive manual annotation to scalable, AI-driven curation strategies guided by expert knowledge.

B. Evolution of Data Types

From a data perspective, experts in EITL workflows contribute domain knowledge to three increasingly abstract forms of supervision: explicit labels, instructions, and comparative preferences.

1) Traditional Supervised Training Data: Supervised data encode explicit input–output mappings for classification, extraction, and multimodal alignment. In specialized domains, experts ensure label validity, while active learning, weak supervision, preliminary annotation, and conflict detection reduce manual cost [34]. These mechanisms are examined in Section IV.

2) Instruction-Following Data: Instruction-following data express task intent through prompts, constraints, examples, and responses. Experts preserve linguistic and domain precision, while guided generation, iterative refinement, and consistency auditing expand coverage [35]. Section V analyzes these mechanisms without attributing distinct workflow components to a single study.

3) Preference Data: Preference data encode comparative judgments over coherence, relevance, safety, and domain consistency. Experts define these latent quality criteria, while selective comparison, model-assisted ranking, and structured review improve scalability without relinquishing semantic authority. Section VI examines their construction and modeling.

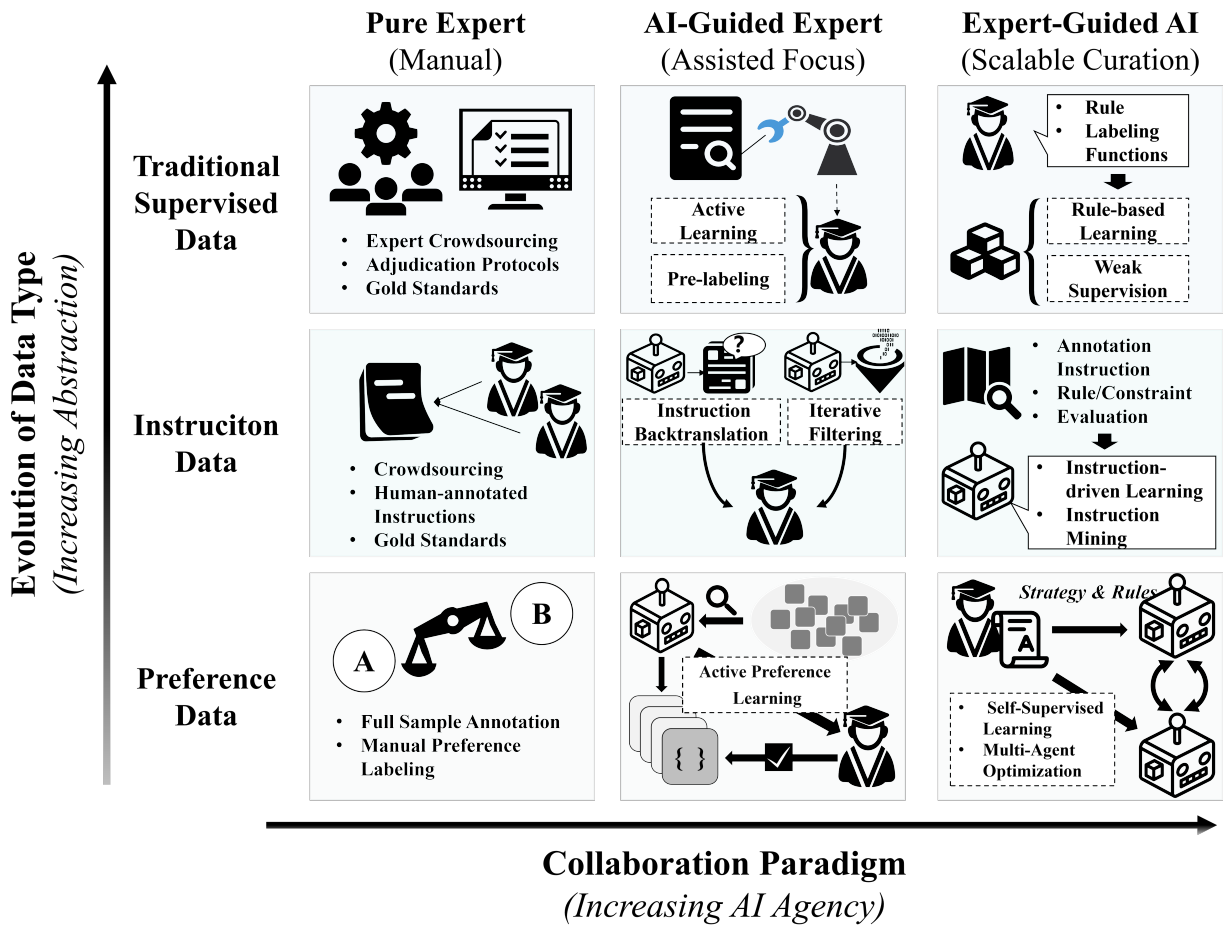


Fig. 3. A taxonomy of expert–AI collaboration paradigms for high-quality training data generation, organized along two orthogonal axes: the vertical axis denotes an analytical progression of data types toward increasing levels of semantic abstraction, while the horizontal axis reflects the degree of AI agency within the collaboration paradigm. Each cell characterizes the dominant methodologies and workflows associated with a specific combination of data type and collaboration modality, illustrating how instance-level production responsibility progressively shifts toward AI systems while experts retain authority through specifications, policies, and auditing.

C. Collaboration Paradigms

For each data type, we compare EITL paradigms across levels of AI agency, showing how experts' roles evolve from direct annotators to governance-oriented supervisors.

1) Pure Expert: In Pure Expert workflows, experts directly produce and adjudicate data. This maximizes semantic authority and fidelity but limits scalability; supporting mechanisms therefore focus on collaboration and arbitration rather than automated data production.

2) AI-Guided Expert: In AI-Guided Expert workflows, AI filters, prioritizes, or proposes samples to concentrate human effort on high-value decisions, while experts remain the primary data producers and final authorities.

3) Expert-Guided AI: In Expert-Guided AI workflows, experts define rules, constraints, or policies, while AI becomes the primary producer of data at scale.

4) A Unified View of Expert Role Evolution: Taken together, these three paradigms reveal a coherent trajectory of expert-role evolution, as shown in Fig. 4:

- in *Pure Expert*, experts are *semantic producers*;
- in *AI-Guided Expert*, experts become *strategic reviewers and curators*;
- in *Expert-Guided AI*, experts act as *rule designers, policy setters, and governors*.

This trajectory does not remove experts from the loop; it relocates their authority from instance-level intervention to schema design, specification, prompt engineering, reward definition, principles, and auditing. The first transition addresses the manual-annotation bottleneck through AI-assisted selection, whereas the second uses capable foundation models for generation, rewriting, ranking, filtering, and partial evaluation under expert governance.

Platform design follows the same progression. Snorkel and Stanford DeepDive encode expert rules for scalable labeling; Prodigy and Argilla route informative samples to human reviewers; and MEGAnno+ combines model generation with human verification. The former Humanloop platform is retained as a historical service that ceased operations in 2025 [36]. Prolific and Toloka can support Pure Expert workflows only when contributors are screened for the required expertise. These examples illustrate how expert-involvement strategies map to platform architectures.

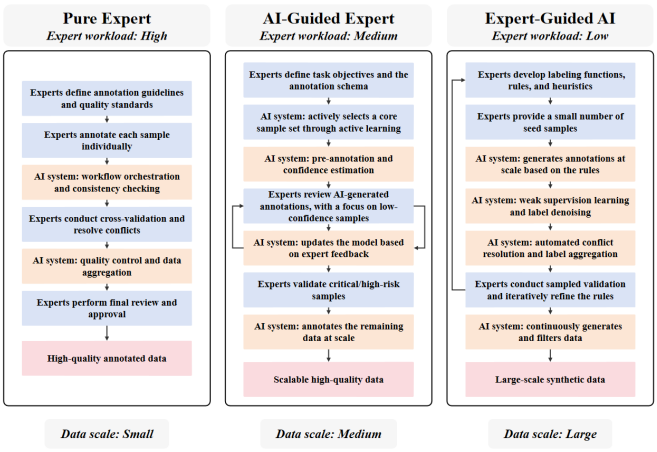


Fig. 4. Evolution of expert–AI collaboration paradigms: from manual annotation to scalable synthesis. The workflow transitions from higher expert workload with small-scale data (left) to lower expert workload with large-scale data (right), demonstrating the shift in expert roles from semantic producers to strategic supervisors and governors.

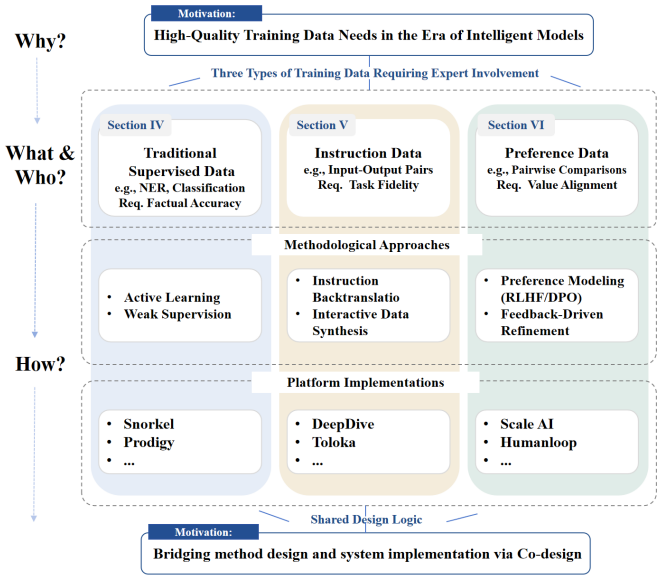


Fig. 5. Organization of this survey.

D. Summary

This survey combines (i) an orthogonal data type × collaboration-paradigm taxonomy and (ii) a methodological and platform-oriented view from algorithm design to deployment.

Fig. 5 illustrates the organization of this survey, and each paradigm reflects a coherent configuration of:

- **Problem structure**: what kind of data is needed.
- **Human–AI division of labor**: who does what.
- **Feedback semantics**: how feedback is interpreted.
- **Platform support**: which platforms operationalize the paradigm.

Together, these dimensions support consistent comparison while accounting for implementation constraints, reproducibility, and deployment relevance.

IV. Traditional Supervised Data Construction with Expert Guidance

A. Core Methods

1) Expert-Centric Adjudication: Large-scale crowd annotation, exemplified by ImageNet, established the value and scalability of collective labeling while exposing variability in annotator competence [37], [38]. Crowdsourcing research more broadly shows that contributor organization and community structure can affect collective innovation performance [39]. To address noise and variability in crowd annotations, GLAD jointly infers annotator competence and item difficulty, automatically upweighting reliable annotators and flagging ambiguous samples [40]. Complementing this approach, Hsueh et al. (2009) proposed filtering criteria based on annotator consistency, inter-annotator agreement, and response time [41]. Although these studies do not assume that all crowd workers are domain experts, their ability-aware aggregation and adjudication mechanisms remain relevant when EITL workflows use experts to establish gold labels or resolve ambiguous cases.

2) Interactive Active Selection for Experts: In AI-guided high-quality dataset construction, active learning and label optimization are pivotal [42]. Sener and Savarese (2018) formulated active learning as a core-set selection problem, focusing on identifying representative subsets through geometric coverage [43]. Yoo and Kweon (2019) introduced a task-agnostic loss-prediction module for estimating the informativeness of unlabeled samples [44]. Confident Learning identifies probable label errors [45], Co-teaching uses two networks to cross-select clean samples [46], and MentorNet learns curriculum weights that prioritize samples likely to be correct [47]. Recent LLM/LMM systems extend this routing principle. Mixture of LLMs in the Loop uses model disagreement to identify unreliable labels, although its original implementation automates label assessment rather than requiring expert annotation [48]. ActiveLLM addresses cold-start active annotation by selecting useful examples before a strong task model is available [49]. ACT flags potentially erroneous multimodal annotations for targeted human review [50], while a human–LMM framework assigns object selection to humans and semantic-label generation to the model [51]. Wu et al. (2025) further combine hierarchical dependencies and class-balanced sampling to query informative multilabel images [52]. Recent user studies indicate that LLM assistants can accelerate specialist annotation but do not eliminate the need for independent expert judgment, and that reusable domain guidelines improve adherence to specialized conventions [53], [54].

3) Expert-Defined Programmatic Weak Supervision: In the construction of high-quality datasets, "Expert-Guided AI" leverages weak supervision and programmatic labeling to turn human expertise into scalable data production. Ratner et al. (2016) introduced the "Data Programming" paradigm, enabling experts to encode domain heuristics into noisy labeling functions, which are then denoised by a generative model to create large training sets [55]. Building on this, the Snorkel system provides a practical framework for experts to programmatically guide AI training through diverse labeling functions, bypassing manual per-sample annotation [56]. Rat-

ner et al. (2019) extended this approach to multitask weak supervision [57], while Snuba automatically induces heuristics from a small gold set [58]. Related semi-supervised methods exploit unlabeled and evolving text streams, but they are enabling learning techniques rather than programmatic weak-supervision workflows [59]. Automated labeling and rationale-generation methods can likewise support an EITL workflow without including experts in their original implementations. CanDist generates candidate labels and then distills them into a single label through a teacher–student framework [60]; RISE generates and self-verifies visually grounded reasoning traces through reinforcement learning [61]. When these methods are embedded in EITL workflows, experts may define task schemas, label spaces, trusted anchors, or audit criteria.

Traditional supervised data construction has evolved from direct human labeling to AI-assisted selection and expert-governed automation. Active selection improves efficiency but may bias coverage, whereas programmatic weak supervision scales expert rules but remains limited by their expressiveness. LLM- and LMM-assisted annotation can reduce expert workload, yet model errors and limited transparency require targeted review. Other supporting techniques include conditional generative models for structured-output synthesis [62], pretrained generative priors such as Marigold that reduce task-specific supervision in dense vision tasks [63], and interpretable multimodal fusion in applications requiring expert validation [64]. The paradigm assignments used below therefore describe how a method functions when embedded in an EITL workflow; some cited methods are enabling techniques whose original implementations do not directly include experts.

B. Platform Support

Snorkel (Ratner et al., 2017) pioneered the "data programming" paradigm for combining weak-supervision sources in rapid, end-to-end data labeling [56]. Building on this approach, Snorkel DryBell (Malkin et al., 2019) demonstrated its viability at industrial scale by using existing organizational knowledge to reduce labeling costs [65].

Dobbie et al. developed Markup [66], a platform that leverages active learning and automated ontology mapping for rapid unstructured-text annotation. Jelenić et al. introduced ALANNO [67], which addresses practical challenges such as data imbalance and multi-annotator collaboration through performance forecasting and quality control. Kim et al. presented MEGAnno+ [68], a human–LLM collaborative system that uses large language models to accelerate labeling while maintaining reliability through human verification. AnnoHID similarly combines LLM preannotation with human validation in an interactive annotation workflow [69].

AutoDive [70] supports scientific-literature annotation through an "on-site" workflow that allows researchers to annotate text directly in PDF documents while preserving layout context and managing projects, resources, and ontologies. Building on this system, AutoDive+ [71] extends the workflow to multimodal elements such as images and tables.

Prodigy [72], developed by Explosion AI, is an extensible annotation platform that uses active learning to prioritize high-value samples. It supports cross-domain task customization and integration with widely used machine-learning tools to reduce annotation workload and accelerate model iteration.

V. Instruction-Following Data Synthesis through Expert–AI Collaboration

A. Core Methods

1) Human-Authored Instruction Curation: Early instruction alignment relied on human-authored supervision in which qualified annotators wrote instructions or demonstrations and adjudicated disagreements; models could still generate candidate responses for comparison.

In instruction and preference data construction, Ouyang et al. (2022) established the foundational role of human-authored supervision in instruction alignment in *Training Language Models to Follow Instructions with Human Feedback* [73]. Qualified human labelers wrote prompts and demonstrations, ranked model-generated responses, and followed carefully designed annotation protocols. This paradigm enables explicit preference modeling and high-fidelity supervision, but its strong dependence on large-scale human labor fundamentally limits scalability and iteration efficiency.

At the level of structured instruction abstraction, Wang et al. (2022) introduced Super-NaturalInstructions, further demonstrating the value of Pure Expert annotation for task generalization [74]. By systematically curating declarative instructions for over 1,600 natural language processing (NLP) tasks—with explicit task definitions, input–output specifications, and examples—the dataset treats instructions as structured carriers of task semantics rather than free-form prompts. While this design significantly enhances cross-task generalization, it also underscores the high cost and limited scalability of purely manual instruction construction.

2) Augmented Instruction Evolution: In AI-Guided Expert pipelines, LLMs generate, evaluate, and filter candidate instructions, while experts may retain final curation authority when these methods are embedded in EITL workflows. This stage combines instruction backtranslation with iterative quality filtering rather than relying on one-shot synthesis; several source methods automate these operations without experts in their original implementations.

Traditional instruction synthesis methods typically follow a forward-generation paradigm in which complex instructions or constraints are constructed first. The model is then required to generate responses conditioned on these instructions and constraints. However, for complex instruction-following tasks, this paradigm heavily depends on the model's ability to faithfully execute intricate constraints, often resulting in noisy synthetic data with unstable quality. To address this limitation, Li et al. (2023) proposed and formalized Instruction Backtranslation as a core framework for large language model self-alignment [75]. Instead of generating responses from instructions, this paradigm constructs training data by reverse-generating instructions from high-quality responses, thereby forming instruction–response pairs. Concretely, a small set of human-annotated instruction–response examples is used to fine-tune a base model into a backward model, which is

then applied to large-scale unlabeled corpora to infer potential instructions from existing responses, enabling bootstrapped data expansion. By shifting the generation difficulty from executing complex instructions to identifying the instructional intent implicit in a response, this paradigm reduces the capabilities required of the model while improving the diversity and naturalness of the generated instructions. Thus, instructional intent is inferred from existing responses rather than being explicitly authored for every example.

Building upon this framework, Qi et al. (2025) further proposed Constraint Back-translation to enhance instruction synthesis for complex instruction-following scenarios [76]. Their key observation is that responses in high-quality instruction datasets often implicitly satisfy several complex constraints. Rather than requiring models to generate responses that satisfy explicit constraints, Constraint Back-translation reverses the process by prompting models to identify and explicitly articulate the constraints already manifested in a response. By transforming constraint execution into constraint recognition, this method significantly improves the fidelity and stability of synthesized data, further strengthening the applicability of instruction backtranslation in complex settings. Within this framework, compositional instruction augmentation can be viewed as a structured complement to instruction backtranslation. For example, Mosaic-IT [77] automatically synthesizes complex multi-instruction training samples by randomly concatenating multiple instruction–response pairs and employing meta-instructions to control output format, order, and selective omission. Notably, this approach requires neither additional human annotation nor teacher models, achieving substantial gains in training efficiency. It is important to note, however, that such compositional methods do not introduce new semantic supervision; instead, they reorganize high-quality instruction data into more complex structures and are therefore best positioned as a downstream structural enhancement rather than an independent generation paradigm.

Despite the strong generative capabilities enabled by instruction backtranslation, data-quality assurance in the AI-Guided Expert paradigm can combine iterative filtering and model-based self-evaluation with expert-defined criteria or selective review. Nguyen et al. (2024) proposed Instruction Back-and-Forth Translation, which introduces a response-rewriting step into the backtranslation pipeline [78]. This method first derives instructions from web-scale textual responses and subsequently employs an aligned LLM to rewrite the original responses conditioned on the generated instructions, adding quality-refinement and normalization constraints to the data-construction process. More broadly, a growing body of work treats large language models as evaluators that score, filter, and iteratively refine generated data. As enabling techniques rather than natively expert-involved pipelines, Self-Refine [79] leverages self-feedback to perform multiround output refinement, while Tuna [80] uses AI-generated feedback to rank candidate responses and improve instruction tuning. In parallel, Zheng et al. (2023) systematically examined the reliability of LLM-as-a-Judge using MT-Bench and Chatbot Arena, providing empirical evidence that supports the use of models as evaluators in iterative filtering pipelines [81].

3) Schema-Driven Instruction Synthesis: As instruction tuning has become a central paradigm for aligning large language models, research has increasingly shifted expert guidance from explicit human annotation toward automated instruction-generation and selection mechanisms that internalize expert behavior within the model. Wang et al. (2023) introduced Self-Instruct, which demonstrated a bootstrapping approach for instruction-data generation using pretrained language models. Starting from a small set of human-authored seed tasks, Self-Instruct iteratively expands diverse instruction–input–output triples, substantially reducing reliance on manually curated instruction data while achieving performance comparable to that of human-aligned models across multiple benchmarks [82].

Subsequently, Honovich et al. (2023) proposed Unnatural Instructions, which further minimizes human involvement by automatically constructing diverse instruction-following data through staged generation and instruction paraphrasing, illustrating the feasibility of replacing explicit expert design with generative mechanisms [83]. To increase instruction complexity, Xu et al. (2024) introduced the Evol-Instruct method, which systematically evolves initial instructions along both depth and breadth dimensions to progressively generate complex open-domain instruction data. Models trained on such data, exemplified by WizardLM, significantly outperform existing open-source models on complex instruction-following tasks [84]. Xu et al. (2023) proposed Baize, which leverages self-chat data generation to simulate expert behaviors in multi-turn conversations, significantly reducing human involvement while maintaining high-quality dialogue generation with minimal parameter adjustments [85]. The knowledge-graph-guided DeepDive method from THUDM automatically synthesizes challenging QA pairs and search trajectories and is therefore treated here as an enabling workflow rather than a natively expert-involved system [86]; it should not be confused with the Stanford DeepDive knowledge-base-construction platform [87].

As the scale of automatically generated instruction data continues to expand, research attention has increasingly shifted toward data efficiency and quality control. LaMini-LM, proposed by Wu et al. (2024), validated the effectiveness of large-scale instruction distillation for training lightweight models [88], while Instruction Mining focused on selecting high-quality instructions from existing data pools using interpretable quality rules, leading to substantial improvements in instruction tuning without requiring additional data generation [89]. From a unifying perspective, Lou et al. (2024) characterized these approaches under the paradigm of indirect supervision, arguing that implicitly injecting expert knowledge into training data through generation, distillation, and selection constitutes a key driver of progress in instruction-following capabilities. They further emphasized dual scaling—the coordinated expansion of model capacity and data scale—as a fundamental trend shaping this research direction [90]. Overall, these methods illustrate a systematic evolution of expert guidance in instruction alignment, from explicit human-designed supervision toward model-driven data construction and selection mechanisms.

The three paradigms expose a fidelity–scale trade-off. Human-authored data provide strong grounding but limited volume; model-assisted generation expands coverage but requires filtering; and autonomous synthesis may propagate latent errors. Reliable instruction construction therefore combines structured expert specifications, scalable generation, and robust evaluation to reduce hallucination and improve performance on complex and out-of-distribution tasks.

B. Platform Support

Pure Expert instruction construction is supported by annotation-centric platforms such as Scale AI and Label Studio [91], [92], which provide interfaces for authoring prompts and responses, reviewing instruction–response pairs, and adjudicating disagreements. These workflows align with human-feedback pipelines in which qualified annotators retain authority over data abstraction, validation, and value alignment [73].

AI-Guided Expert platforms combine model APIs for candidate generation with model-based evaluation and selective human review. LLM-as-a-Judge frameworks support scalable scoring [81], while SageMaker Ground Truth Plus manages demonstrations and preference comparisons [93]. Findings that aggregated nonexpert annotations can approach expert labels concern human annotation cost and quality rather than model-based evaluation [94]; expert qualification therefore remains task dependent.

Expert-Guided AI platforms operationalize expert knowledge through schemas, constraints, or policies that govern automated synthesis and selection. Declarative systems such as Stanford DeepDive illustrate rule-based execution and joint inference [87], although it is an enabling architecture rather than a native instruction-generation platform. The former Humanloop platform historically supported iterative LLM feedback but is retained only as a historical platform that ceased operations in 2025 [36]. Across these designs, expert work shifts from instance authoring toward supervision and governance.

VI. Preference Data Collection and Modeling with Expert Input

A. Core Methods

1) Expert-Driven Preference Data Generation: Expert-driven preference data generation refers to systems in which experts or highly trained annotators directly perform preference judgments at the sample level. Intelligent components in such systems are primarily responsible for workflow orchestration, consistency checking, and quality control, rather than preference inference. From a human–AI systems standpoint, this paradigm maximizes human authority and interpretability by ensuring that preference semantics are explicitly defined and validated by experts. Consequently, expert-driven approaches are particularly suitable for safety-critical, high-stakes, or domain-sensitive applications. However, the reliance on expert labor imposes significant constraints on scalability and cost efficiency.

Ouyang et al. established this paradigm by formalizing preference learning through explicit human pairwise comparisons and reward modeling [73]. In this framework, preference labels originate from human annotators, while models generate candidate responses and learn from the resulting judgments. *Improving Alignment of Dialogue Agents via Targeted Human Judgements* decomposes desired dialogue behavior into natural-language rules and elicits rule-specific human judgments, enabling more efficient rule-conditional reward modeling [95].

Subsequent datasets such as HelpSteer3-Preference extend expert-driven preference construction across tasks and languages, emphasizing annotation guidelines, inter-annotator agreement, and semantic consistency as system-level quality objectives [96]. OpenAssistant Conversations represents a large-scale instantiation of this paradigm through community-based human annotation [97]. Although not all annotators are domain specialists, preference labels are still produced through direct human comparison and subjected to structured review processes, maintaining alignment with expert-driven system-design principles.

2) Human-Centered, AI-Assisted Preference Data Generation: Human-centered, AI-assisted preference data generation characterizes systems in which experts retain authority over preference semantics, while intelligent components actively assist in candidate generation, uncertainty estimation, or selective sampling. Instead of exhaustively labeling all data, experts intervene at strategically selected decision points, such as ambiguous, safety-critical, or low-confidence cases. This paradigm embodies a closed-loop human–AI system in which expert feedback continuously shapes model behavior, enabling improved scalability without relinquishing semantic control.

Skywork-Reward-V2 provides a prototypical example through a two-stage curation pipeline: humans establish verified gold data, after which LLMs and reward models scale automatic curation under human guidance [98]. PKU-SafeRLHF combines answers generated by Llama-family models with human annotations of safety meta-labels and helpfulness/harmlessness preferences [99].

These two systems illustrate the defining boundary of the AI-Guided Expert paradigm: models generate or organize candidates, but humans still provide the consequential safety labels, preference judgments, or targeted validation. Methods that derive most preference signals automatically from reward models, task metrics, principles, or self-generated evaluations are discussed in the following Expert-Guided AI category.

3) AI-Driven Preference Data Generation with Expert Oversight: AI-driven preference data generation with expert oversight describes systems in which preference labels are predominantly generated by intelligent agents, with human experts contributing indirectly through system design, constraints, or policy specification. From a systems engineering perspective, this paradigm shifts expert involvement from operational execution to high-level governance, enabling continuous and large-scale preference generation. While this approach offers significant scalability advantages, it introduces new challenges related to preference drift, self-reinforcement, and reduced transparency.

Finding the Sweet Spot selects response pairs using reward-model signals, Self-Preference derives preference pairs from task metrics, and SAFER-INSTRUCT synthesizes safety-oriented preference comparisons without human annotators [100]–[102]. These methods can be embedded in EITL workflows when experts define objectives, safety principles, reference data, or audit criteria, but the source methods should not be described as direct human–AI annotation pipelines. UltraFeedback positions models as both generators and evaluators [103], while Constitutional AI encodes human-authored normative principles that guide AI self-critique and preference generation [104].

Building upon these foundations, Learning LLM-as-a-Judge for Preference Alignment trains models to act as autonomous evaluators [105]. Spread Preference Annotation begins with a small amount of human preference data and progressively generates self-annotated preferences [106]; ICON^2 stabilizes self-synthetic preference learning through intrinsic regulation [107], and heterogeneous-consistency co-alignment introduces cross-model consistency signals for automated preference learning [108]. Contrastive Preference Learning [109] and sequence-level preference optimization [110] consume offline or automatically constructed rankings as training signals; they optimize from preference data rather than generating the original judgments.

Weak or noisy preference signals can exacerbate hallucination and unreliable generation, particularly in domain-specific and safety-critical settings. Expert-in-the-Loop preference construction can help mitigate these problems by introducing verified expert judgments into preference pairs and reward signals. In the context of instruction tuning and preference optimization, such expert supervision improves factual grounding and reduces inconsistent outputs. It also enhances robustness under distribution shift by stabilizing preference definitions across annotators and tasks.

We compare the three paradigms across annotation cost, expert workload, noise robustness, alignment quality, and application scenarios. Expert-driven methods are best suited to safety-critical tasks that require high semantic precision, but they scale poorly and incur high costs. AI-driven methods are suited to large-scale, low-risk settings, offering high efficiency but weaker stability and a higher risk of preference drift. Human-centered hybrid methods are effective in medium-risk and iterative-development settings, but their performance depends on the quality of expert feedback and sampling design. These trade-offs suggest that expert-in-the-loop systems should be designed in a scenario-aware and risk-adaptive manner rather than relying on a fixed collaboration mode.

B. Platform Support

Expert-Driven platforms primarily orchestrate direct judgments, reviewer management, and quality assurance. Scale AI, Labelbox, and Toloka provide managed annotation or configurable comparison workflows [91], [111], [112]; OpenTrain AI recruits AI trainers [113], whereas Prolific recruits screened research participants [114]. Because recruitment services do not themselves guarantee domain expertise, project designers must qualify reviewers, who retain semantic authority over the resulting preferences.

Human-centered, AI-assisted platforms combine candidate generation with selective review. Argilla and Label Studio support configurable feedback datasets and model-assisted inspection [92], [115], while SageMaker Ground Truth Plus manages supervised-fine-tuning demonstrations and preference comparisons [93]. Scale AI can support similar managed workflows. The former Humanloop platform is included only as historical context because its official site documents its acquisition and 2025 sunset but does not provide sufficient evidence to reconstruct all of its former capabilities [36]. Across these systems, qualified reviewers retain authority over consequential judgments.

For AI-driven preference generation, Sama AI, Scale AI, and Surge AI can be configured within workflows in which models curate candidates, rank preferences, or validate outputs under expert-defined constraints [91], [116], [117]. Such workflows may include reward-model training and automated validation. Human experts specify policies, quality thresholds, and audit checkpoints rather than labeling every sample. This division reduces direct labor and improves scale while preserving process-level oversight of preference drift, ethical compliance, and final acceptance.

VII. Case Study

To illustrate how Expert-in-the-Loop systems support high-quality data construction through microtask decomposition, we summarize a real-world workflow built around AutoDive+ [71] for data preparation in the domain of microbe-enabled green materials. First, during *ontology definition*, experts specify entity types, relations, annotation rules, dictionaries, and labeling guidelines. Second, during *preliminary-label verification and NER annotation*, the system applies these resources and model-assisted labeling, and experts review and correct the outputs. Third, verified entities and relations are organized into a domain knowledge graph, with model assistance used to identify deficiencies and experts validating the resulting knowledge. Finally, during *disagreement adjudication and final validation*, annotators review candidate records, while experts resolve conflicts and feed their decisions back into guideline updates. Thus, structured outputs from each stage become inputs to the next, illustrating a transition from AI-Guided Expert assistance to Expert-Guided AI governance while enabling scalable construction with expert quality control. The resulting annotated knowledge supported knowledge-graph construction and model training in MicrobeDiscover [118]; detailed procedures, interfaces, and examples are available at https://github.com/kg4sci/awesome-EITL.

VIII. Challenges, Future Trends, and Conclusion

Despite rapid progress in Expert-in-the-Loop (EITL) methods and platforms, the scalable, reliable, and domain-grounded production of high-quality training data remains hindered by five persistent challenges.

In Tables I and II, we categorize the intelligent methods and data-construction tools discussed in the surveyed literature.

TABLE I
COMPARATIVE SYNTHESIS OF METHODS USED IN OR ENABLING EXPERT-INVOLVED INTELLIGENT DATA CONSTRUCTION.

Data Type	Paradigm	Method Family	Representative Work	Expert–AI Roles and Trade-off
Supervised Data	Pure Expert	Collective human labeling and adjudication	ImageNet; crowd-label learning; GLAD; annotation-selection criteria [37], [38], [40], [41]	Human annotators label; experts may establish gold standards or resolve ambiguity. AI aggregates. High fidelity, but costly and sensitive to worker quality.
	AI-Guided Expert	Coverage- and loss-based selection	Granular, core-set, loss-prediction, and hierarchical active learning [42]–[44], [52]	AI selects; experts label. Efficient, but selection may bias coverage.
		Label-error detection and robust learning	Confident Learning; Co-teaching; MentorNet [45]–[47]	AI flags or weights samples; experts inspect or relabel selected cases. Robust to noise, but prediction-dependent.
		LLM/LMM-assisted annotation	MoLAM; ActiveLLM; ACT; human–LMM; expert-workflow studies [48]–[51], [53], [54]	AI proposes or prioritizes; experts may audit when these methods are incorporated into EITL. Scalable, but exposed to model errors.
	Expert-Guided AI	Programmatic weak supervision	Data Programming; Snorkel; multitask weak supervision; Snuba [55]–[58]	Experts encode rules; AI applies and denoises them. Scalable, but rule-limited.
		Automated labeling and rationale synthesis	CanDist; RISE [60], [61]	Experts may define schemas, anchors, or audits; AI generates, distills, or self-verifies. Scalable, but expert involvement may be indirect.
Instruction Data	Pure Expert	Human-authored instructions	InstructGPT; Super-NaturalInstructions [73], [74]	Qualified humans author and adjudicate; AI learns and may generate candidate responses. Reliable, but difficult to scale.
	AI-Guided Expert	Backtranslation and constraint recovery	Instruction, constraint, and back-and-forth translation [75], [76], [78]	AI reconstructs and rewrites; experts may curate in EITL use. Diverse, but model-dependent.
		Composition, refinement, and evaluation	Mosaic-IT; Self-Refine; Tuna; LLM-as-a-Judge [77], [79]–[81]	AI composes, filters, and scores; experts may set criteria in EITL use. Efficient, but may amplify bias.
	Expert-Guided AI	Bootstrapping and evolution	Self-Instruct; Unnatural Instructions; WizardLM; Baize [82]–[85]	Human-authored seeds or rules may guide AI expansion. Scalable, but error-prone.
		Distillation, mining, and indirect supervision	LaMini-LM; Instruction Mining; indirect-supervision survey [88]–[90]	Curated sources or views bound AI distillation and selection. Efficient, but source-bounded.
Preference Data	Pure Expert	Direct comparison and review	InstructGPT; targeted judgments; HelpSteer3; OpenAssistant [73], [95]–[97]	Qualified human annotators define and rank preferences. Interpretable, but labor-intensive.
	AI-Guided Expert	Human–AI preference curation	Skywork-Reward-V2; PKU-SafeRLHF [98], [99]	AI generates or curates candidates; humans provide safety labels, preferences, or targeted validation. Balanced, but costly.
	Expert-Guided AI	Automated preference construction and optimization	Sweet Spot; Self-Preference; SAFER-INSTRUCT; UltraFeedback; Constitutional AI; learned judges and propagation; ICON^2; HCC; CPL/CPO [100]–[110]	Humans may supply principles, metrics, references, or audits; AI generates, ranks, propagates, or optimizes. Scalable, but vulnerable to drift and self-reinforcement bias.

Paradigm assignments describe how methods are used in an EITL workflow; some cited methods are enabling techniques and do not include experts in their original implementations. The data-type labels in Table II indicate the primary uses emphasized here rather than exclusive platform capabilities. Table III further introduces complementary evaluation dimensions for systematic benchmarking beyond system design. Relevant publications and platform links are available at https://github.com/kg4sci/awesome-EITL.

A. Limitations and Challenges

First, the *expert bottleneck* persists. Domain experts are scarce, time-constrained, and costly to engage. Current workflows often treat expert input as a linear, monolithic step rather than decomposing tasks into cognitively optimal micro-interventions. This leads to inefficient utilization of expert bandwidth and limits throughput. **Second, *semantic misalignment between tools and expertise* undermines quality.** Many annotation platforms prioritize interface flexibility over domain expressivity: they support bounding boxes or spans but lack native constructs for encoding clinical uncertainty, legal conditionality, or causal reasoning. As a result, experts must encode domain knowledge in rigid schemas, thereby losing nuance and increasing cognitive load. **Third, *conflict resolution remains difficult to scale.*** When multiple experts disagree because of evolving guidelines, edge cases, or interpretive divergence, resolution typically relies on ad hoc discussion, hierarchical arbitration, or majority voting. These processes are slow, poorly documented, and often fail to capture why disagreement occurred. Existing mechanisms provide partial support for recording and aggregating disagreement, but systematic learning from the causes of conflict remains limited. **Fourth, *evaluation of data quality lacks domain-grounded metrics.*** Downstream model performance remains the de facto proxy for data quality, yet it conflates data flaws with model-architecture limitations. We lack standardized, interpretable metrics for domain fidelity, logical consistency, or coverage adequacy. Adams et al. meta-evaluate faithfulness metrics for model-generated long-form hospital-course summaries, while Liu et al. study robust human evaluation of generated summaries; these works concern the evaluation of incorrect, missing, or unsupported generated content rather than errors in expert-authored annotations [119], [120]. Klie et al. systematically analyze annotation-quality management, including common errors in the calculation and interpretation of inter-annotator agreement, annotator management, adjudication, and validation [121]. For structured analysis, ontology-based knowledge graphs can organize AI tasks, benchmarks, and evaluation measures [122], while taxonomies of object-detection annotation errors organize defect types

TABLE II
SUMMARY OF REPRESENTATIVE EITL TOOLS AND PLATFORMS.

Tool or Platform (Primary Data Type)	System Architecture	Scalability	Expert Workflow Integration
Snorkel; DryBell (Supervised) [56], [65]	Probabilistic weak-supervision pipeline with distributed rule execution	Programmatic labeling for large datasets	Experts encode and refine labeling functions.
Markup; ALANNO; Prodigy (Supervised) [66], [67], [72]	Active-learning-assisted annotation and selective sample routing	Prioritization reduces expert review volume	Experts annotate and validate selected cases.
MEGAnno+; AnnoHID; AutoDive+ (Supervised; Instruction-enabling) [68], [69], [71]	Human–LLM annotation with schema, ontology, and multimodal support	Model-assisted data construction across modalities	Experts define prompts or schemas and verify outputs.
DeepDive (Knowledge-base construction; Supervised) [87]	Declarative rules with probabilistic joint inference	Rule-based extraction scales across corpora	Experts encode knowledge as logical constraints.
Scale AI; Labelbox; Surge AI (Multiple data types) [91], [111], [117]	Managed annotation, human-feedback, evaluation, and quality-assurance pipelines	Parallel workflows with enterprise workforces	Qualified annotators or experts label, rank, validate, and adjudicate.
Label Studio; Argilla (Multiple data types) [92], [115]	Configurable labeling and feedback schemas with model-assisted review	Flexible deployment and iterative review	Experts design tasks and refine model suggestions.
SageMaker Ground Truth Plus (Instruction/Preference) [93]	Managed human-feedback workflows for demonstrations and preferences	Managed workforce and repeatable workflows	Qualified reviewers create demonstrations and rank outputs.
Sama; Toloka (Multiple data types) [112], [116]	Managed annotation, evaluation, and human-feedback workflows	Distributed workforces and batch processing	Qualified annotators or experts validate and audit outputs.
OpenTrain AI; Prolific (Preference/Evaluation) [113], [114]	Trainer or participant recruitment and controlled data collection	Vetted marketplaces and study management	Researchers screen contributors and collect direct judgments.

TABLE III
REPRESENTATIVE EVALUATION DIMENSIONS FOR EXPERT-IN-THE-LOOP (EITL) SYSTEMS.

Evaluation Dimension	Description	Representative Metrics	Evaluation Focus
Annotation Quality	Measures the correctness and reliability of expert-generated annotations.	Accuracy, precision, recall, F_1 score	Annotation correctness
Expert Agreement	Evaluates the consistency among multiple experts during annotation or review.	Cohen's κ, Fleiss' κ, Krippendorff's α	Annotation consistency
Expert Workload	Measures the amount of human effort required throughout the annotation process.	Person-hours, interaction count, correction count	Human effort
Annotation Efficiency	Evaluates improvements in annotation speed and resource utilization.	Throughput, cost per accepted annotation, annotation time	Efficiency and scalability
Domain Consistency	Assesses whether annotations conform to domain knowledge and expert expectations.	Expert assessment, domain-specific compliance	Domain reliability
Alignment Effectiveness	Measures the contribution of annotated data to downstream model alignment performance.	Hallucination rate, truthfulness, helpfulness	LLM alignment quality
Privacy and Security	Evaluates risks associated with expert participation and sensitive data handling.	Privacy-leakage tests, attack success rate, compliance assessment	Data governance

and their effects [123]. Beyond post hoc evaluation, recent hybrid-intelligence approaches, such as PAQC [124], integrate physics-aware priors with deep representations to enable automated quality control and high-confidence pseudo-label generation. **Fifth, *privacy-preserving expert collaboration remains underexplored.*** EITL systems also raise significant privacy and security concerns because expert-annotated data often contain sensitive information and are vulnerable to inference attacks. Distributed expert collaboration further enlarges the attack surface across the data life cycle. Emerging techniques such as federated learning and differential privacy offer potential solutions but introduce trade-offs between data utility and protection. Designing privacy-aware EITL frameworks remains an important open challenge.

B. Future Directions

To address these challenges, we identify four converging directions for future research.

1. Microtasking and Cognitive-Load-Aware Workflows. Next-generation annotation platforms are expected to decompose complex expert work into atomic, context-rich microtasks. Examples include "verifying the temporal ordering of events in this surgical note" and "assessing the pharmacological plausibility of this drug–drug interaction." Task assignment can be guided by real-time estimates of cognitive load derived from interaction latency, gaze patterns, or physiological signals. Platforms such as Snorkel Flow and the former Humanloop platform illustrate related selective human–AI workflow patterns, although cognitive-load-aware task assignment remains an open research direction.

> **Relevant Questions:**
> How can we automatically detect declines in expert attention and trigger adaptive task restructuring?
> What is the quantitative relationship between microtask granularity and annotation accuracy?

2. Domain-Native Annotation Schemas.
A second trend involves embedding domain-specific ab-

stractions directly into annotation data models, moving beyond flat categorical labels toward structured, executable knowledge representations. Examples include uncertainty intervals for clinical diagnoses, conditional logic blocks for legal clause interpretation, and causal graphs for safety incident reports. Such representations enable richer downstream reasoning, facilitate formal verification, and support interoperability with domain ontologies. This evolution reflects a growing recognition that annotation schemas must mirror the epistemic complexity of the domains they serve rather than impose reductive simplifications that compromise downstream model fidelity.

> **Relevant Questions:**
> How can clinical uncertainty intervals be formally encoded in annotation schemas?
> How can alignment between domain ontologies and annotation schemas be measured?

3. Conflict-Aware Data Engineering.

A third emerging direction reframes annotator disagreement not as noise to be suppressed but as a first-class epistemic signal warranting systematic analysis. Future frameworks should support conflict-provenance logging, root-cause analysis or clustering, and targeted remediation tasks. This direction can transform inter-annotator conflict from a quality-assurance cost into a catalyst for iterative dataset and guideline refinement.

> **Relevant Questions:**
> Can the root causes of disagreements be automatically classified?
> Are the corresponding conflict-resolution solutions traceable and reproducible?

Representative systems and algorithms provide components for this direction [65]. For instance, Snorkel models and probabilistically aggregates conflicting labeling sources through a generative label model [56]. In parallel, platforms such as Argilla support multiple responses and the inspection of disagreement, enabling practitioners to review inter-annotator divergence and refine annotation policies [115]. These capabilities provide foundations for closed-loop, conflict-aware data improvement.

4. Foundation-Model-Assisted Data Curation.

Finally, LLMs are increasingly deployed as assistants in data-curation workflows. They can retrieve contextually relevant clinical guidelines, summarize expert rationales, draft correction suggestions, and validate logical consistency against established domain ontologies. A promising direction is to use retrieval-augmented generation (RAG) pipelines to amplify expert judgment without supplanting it, thereby preserving the epistemic authority of domain specialists while reducing their operational burden.

> **Relevant Questions:**
> To what extent do LLM-assisted RAG pipelines preserve expert authority?
> Which types of tasks are appropriate for LLM-generated recommendations, and which must remain subject to expert adjudication?

C. Conclusion

The practical deployment of EITL systems highlights several critical human-centered challenges. Heavy cognitive workload and annotation fatigue directly degrade data quality. In parallel, multi-expert collaboration introduces coordination overhead. Beyond operational efficiency, ethical design is fundamental. System developers must actively mitigate data bias and ensure transparency in automated recommendations.

Ultimately, the future of EITL systems is not about removing experts from the loop through automation; it is about designing interaction loops that respect human cognition. The success of next-generation EITL frameworks should not be evaluated solely by annotation throughput. Future evaluations must adopt multidimensional criteria, including the depth of domain grounding, the traceability of curatorial decisions, and the resilience of the resulting data ecosystems under evolving real-world demands.

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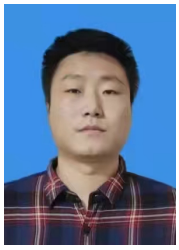
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